A Comparative Study of CNN and Vision Transformer Architectures for Multi-Class Alzheimer's Disease Classification Using MRI Images

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Abstract. Alzheimer's disease (AD) is a progressive neurodegenerative disease affecting memory and cognitive function. Early diagnosis is important for effective treatment and management of disease. In this study, we compare the performance of Convolutional Neural Network (CNN) and Vision Transformer (ViT) architectures for multi-class Alzheimer's disease classification based on MRI images .

The current work evaluates the performance of MobileNetV2, EfficientNet-B0, Res-Net50, DenseNet121 and Vision Transformer models with same experimental setup. The models are assessed with respect to accuracy, precision, recall and F1-score as well as computational efficiency measures such as inference time and model size.

The experimental results show that the Vision Transformer has the best classification performance, because it is able to capture the global contextual relations through the self-attention mechanisms. DenseNet121 and ResNet50 also showed strong performance and MobileNetV2 was better in terms of computational efficiency. Attention-based visualization offers additional insight into the clinically relevant brain regions associated with Alzheimer's disease progression.

The results emphasize the performance of transformer-based architectures for the automated classification of Alzheimer's disease and provide insights into the trade-off between performance and computational efficiency of deep learning models.

Keywords: Alzheimer's Disease · Vision Transformer · CNN · MRI Classification · Deep Learning

1 Introduction

Alzheimer's disease (AD) is a progressive neurological disease that impairs everyday functioning, memory, and cognitive ability. It mostly affects the elderly and is one of the main causes of dementia globally. Alzheimer's disease must be diagnosed early in order to design a proper course of therapy and reduce the illness's progression.

Hippocampal atrophy, ventricular enlargement, and cortical thinning are among the structural brain abnormalities linked to Alzheimer's disease that are frequently detected by magnetic resonance imaging (MRI). These biomarkers offer crucial details about the course of the illness. However, manually interpreting MRI images takes a lot of time and varies from radiologist to radiologist.

Automated medical image analysis has been greatly enhanced by recent developments in deep learning and artificial intelligence. When it comes to extracting hierarchical characteristics from MRI scans for illness classification, Convolutional Neural Networks (CNNs) have demonstrated excellent performance. Medical imaging applications have successfully used architectures including MobileNetV2, EfficientNet-B0, ResNet50, and DenseNet121.

Transformer-based models have lately drawn a lot of interest in computer vision and medical image analysis, in addition to CNN-based designs. Vision Transformer (ViT) captures global contextual links between visual patches by using self-attention techniques. ViT is better at modeling long-range dependencies and spatial interactions than CNNs, which mostly concentrate on local feature extraction.

The CNN and Vision Transformer topologies for multi-class Alzheimer's disease classification using MRI images are compared in this work. MobileNetV2, EfficientNet-B0, ResNet50, DenseNet121, and Vision Transformer are among the models that were chosen. To guarantee a fair comparison, the models are assessed under equal experimental conditions.

The major contributions of this study are summarized as follows:

- CNN and Vision Transformer topologies are compared for Alzheimer's disease categorization.
- F1-score, accuracy, precision, and recall are used in performance analysis.
- FLOPs, inference time, and model size are all included in the computational efficiency study.

2 Literature Review

Memory, cognitive function, and behavioral abilities are all impacted by Alzheimer's disease (AD), a progressive neurological condition. Effective treatment planning and disease management depend on an early diagnosis of AD. Using neuroimaging methods like magnetic resonance imaging (MRI), researchers have investigated a number of computational strategies for automated Alzheimer's disease detection during the last 20 years.

Alzheimer's disease categorization research was first dominated by traditional machine learning techniques. These approaches used classifiers like Support Vector Machines (SVM), k-Nearest Neighbors (k-NN), and Random Forests in conjunction with manually developed feature extraction techniques. For classification tasks, structural brain parameters such as hippocampus volume, cortical thickness, and ventricular enlargement were often employed [30,34]. These methods were somewhat successful, but their effectiveness was mostly dependent on expert knowledge and manual feature engineering.

By making it possible to automatically extract features from unprocessed picture data, deep learning dramatically changed medical image analysis. Convolutional Neural Networks (CNNs), in particular, have shown impressive performance gains in a variety of computer vision applications [1, 2].The capacity of CNN architectures to directly learn hierarchical feature representations from MRI images led to their widespread use in medical imaging [1,2,9].

Among the first CNN architectures to be effectively used for image classification tasks were AlexNet and VGGNet [2, 21]. Deeper architectures like GoogLeNet later offered enhanced feature extraction capabilities via inception modules [22]. By adding residual learning and skip connections, ResNet further addressed vanishing gradient issues and made it possible to train very deep neural networks effectively [3]. DenseNet reduced parameter redundancy and enhanced classification performance by increasing feature propagation and reuse through dense connection across layers [4].

CNN designs have been used in a number of studies to diagnose Alzheimer's disease using MRI data. DeepAD, a deep convolutional neural network architecture for Alzheimer's disease classification utilizing MRI and fMRI data, was proposed by Sarraf and Tofighi [17]. Using MRI scans, Basaia et al. created a deep learning framework for the automated categorization of moderate cognitive impairment and Alzheimer's disease [32]. Wen et al. showed the value of deep learning in neuroimaging applications and offered a thorough review of CNN-based methods for Alzheimer's disease classification [19].

Lightweight architectures like MobileNetV2 and EfficientNet were created to solve the computational difficulty of deep CNN models. MobileNetV2 maintains competitive performance while lowering the number of parameters and computational cost by using inverted residual blocks and depthwise separable convolutions [5]. Compound scaling was established by EfficientNet to systematically balance network breadth, depth, and picture resolution [6]. These lightweight designs are especially crucial for use in healthcare settings with limited resources.

Medical image processing has been substantially enhanced by recent developments in transformer-based systems. For image classification tasks, Vision Transformer (ViT) uses transformer encoder blocks with self-attention mechanisms [7,8]. ViT models are able to capture long-range connections between picture patches and global contextual linkages, in contrast to CNNs, which mainly concentrate on local feature extraction.

Multi-Head Self-Attention (MHSA) layers and feed-forward neural networks make up the Transformer Encoder Block. By enabling the model to discover connections between remote areas of the picture, the self-attention technique enhances global feature representation [7]. Touvron et al. used attention-based distillation approaches to further increase transformer training efficiency [20].

Recently, transformer-based systems have shown encouraging results in applications related to medical imaging. Hybrid CNN-transformer architectures were suggested by Zhou et al. for medical picture categorization [14]. Swin UNETR, a transformer-based framework for brain tumor segmentation in MRI images, was presented by Hatamizadeh et al. [28]. These findings suggest that dispersed structural anomalies linked to neurological disorders can be accurately captured by transformer-based models.

Deep learning architectures for Alzheimer's disease diagnosis have been assessed in a number of comparative studies. For MRI-based Alzheimer's disease diagnosis, Islam and Zhang introduced an ensemble CNN framework and showed better classification performance [16]. In neuroimaging applications, Abrol et al. shown that deep learning models can perform better than conventional machine learning methods [15]. Nevertheless, the majority of current research mostly concentrates on CNN-based designs and offers little comparisons with transformer-based models.

Explainability and interpretability in medical imaging constitute another significant area of research. Clinically significant brain areas impacting model predictions may be found using saliency maps and attention-based visualization approaches [13]. Explainable AI techniques increase reliability and make it easier to incorporate deep learning systems into frameworks for clinical decision assistance.

There are still a number of issues with the current research, despite tremendous advancements in deep learning-based Alzheimer's disease diagnosis. Instead of systematically comparing several CNN and transformer-based models, the majority of research concentrates on particular designs. Despite their significance in real-world healthcare systems, computational efficiency, inference time, and deployment viability are frequently disregarded.

Furthermore, little research has examined how well Vision Transformer models and lightweight CNN architectures perform in multi-class Alzheimer's disease classification under the same experimental conditions. This paper offers a thorough comparison of MobileNetV2, EfficientNet-B0, ResNet50, DenseNet121, and Vision Transformer designs utilizing MRI pictures in order to fill these research gaps. To find appropriate designs for automated Alzheimer's disease diagnosis, the models are assessed in terms of classification performance, computational efficiency, and interpretability.

3 Dataset and Preprocessing

Brain magnetic resonance pictures classified into four classes, non demented, very mild dementia, mild dementia, and moderated dementia, comprise the data set used in this study. The Alzheimer's Disease Neuroimaging Initiative and other publicly accessible medical imaging sources are the source of the dataset.

Anatomical changes linked to neurodegeneration are highlighted in each MRI scan, which contains structural brain information. To provide impartial training, the dataset has a balanced distribution among the various classes.

The photos are downsized to a consistent resolution of 224×224 pixels after being preprocessed to eliminate noise. Rotation, flipping, and scaling are examples of data augmentation techniques that are used to increase model generalization and diversify datasets.

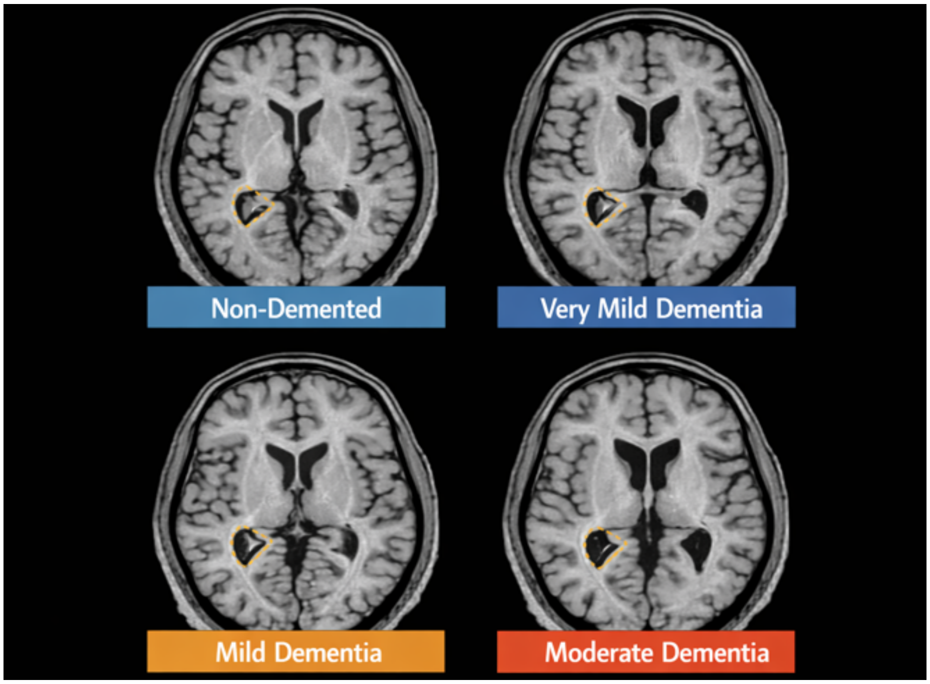

Fig. 1. Four-class of Alzheimer's disease.

To ensure fair comparison, the same dataset and preprocessing pipeline were used for all evaluated models, including MobileNetV2, EfficientNet-B0, ResNet50, DenseNet121, and Vision Transformer.

3.1 Experimental Setup

The research was conducted in a GPU-enabled environment using the TensorFlow and Keras frameworks. The model was trained at a learning rate of 1×10^{-4} using the Adam optimizer.

A batch size of 32 was used for 50 training epochs. Early stopping was employed to prevent overfitting, and the model with the best validation accuracy was selected.

The dataset was divided into training, testing, and validation sets using an 80—10—10 ratio.

3.2 Performance Assessment Criteria

Several statistical metrics are taken into consideration in order to assess the efficacy of the suggested deep learning models. These metrics aid in assessing the models' capacity for categorization throughout various phases of Alzheimer's disease.

The percentage of properly predicted samples over all instances is known as classification accuracy. Furthermore, accuracy is used to gauge how accurate the model's positive predictions are.

Recall is used to assess the model's accuracy in identifying pertinent illness categories, especially when it comes to differentiating between various stages of dementia. Additionally, the F1-score is regarded as a balanced statistic that integrates recall and accuracy into a single performance indicator. When combined, these evaluation indicators provide a reliable assessment of the model's robustness, prediction consistency, and overall classification performance.

To ensure fair comparison, the same dataset and preprocessing pipeline were used for all evaluated models, including MobileNetV2, EfficientNet-B0, ResNet50, DenseNet121, and Vision Transformer.

3.3 Image Resizing and Normalization

All MRI images were reduced to $224 * 224$ pixels in order to fit the Vision Transformer's input size.After scaling, the pixel values were normalized to the $[0, 1]$range to enhance training and reduce scanner-related variability.

3.4 Data Augmentation

During training, a number of augmentation approaches were used to boost the dataset's variety and decrease overfitting. By simulating natural fluctuations in MRI acquisition,These changes improve the model's ability to generalize to new data. The following transformations were used:

- **Rotation at Random:**Pictures were rotated within a narrow range of angles(up to $\pm 20°$) to account for differences in head orientation.
- **Horizontal Flipping:** Occasionally, reflections were used to add variations in left-right symmetry.
- **Zooming:** Zoom-in and zoom-out transformations were applied within the 0.8–1.2 range to simulate differences in scanning distance.
- **Width and Height Shifts:** To prevent overfitting to certain pixel alignments, small spatial translations were added.
- **Light Contrast Adjustments:** To simulate fluctuations in MRI scanner conditions, slight brightness and contrast adjustments were made.

To guarantee consistency between the test and validation sets for an objective assessment, these augmentations were only applied to the training set.

3.5 Train, Validation, and Test Split

Three different subsets have been created from the initial dataset: Training – 80% Validation – 10% Test – 10% Such division ensured that each of the three subsets contained equal proportions of all classes in the data, as it was based on stratified sampling.

3.6 Preprocessing Pipeline

Image capture, scaling, normalization, augmentation, dataset separation, and model input creation are all included in the preprocessing framework. The preprocessing pipeline utilized in this investigation is shown in Figure 2.

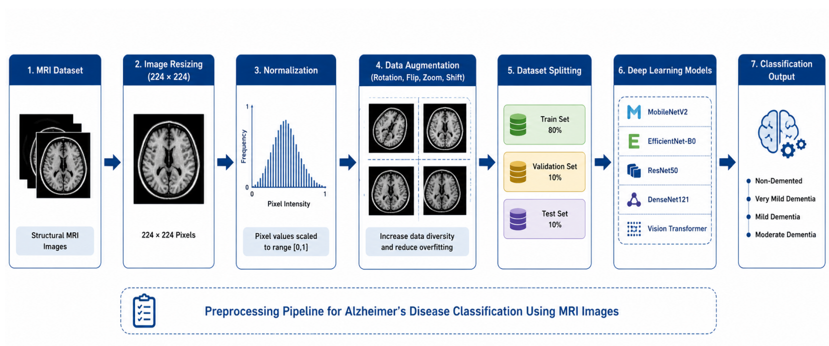

Fig. 2. MRI image preprocessing pipeline for Alzheimer's disease classification.

3.7 Challenges in MRI Data Processing

The data analysis revealed several challenges that needed to be addressed in later stages:

- **Class imbalance**, especially in the ModerateDemented category.
- **Noise and variation** from different MRI machines.
- **Overlapping visual features** between VeryMild and Mild dementia cases.
- **Subtle structural variations** that require a model capable of long-range attention.

The Vision Transformer model, as well as the decisions made regarding preprocessing, augmentation, and model design, were influenced by these discoveries.

4 Methodology

4.1 Overall Framework

Image preprocessing, feature extraction, model training, classification, and performance evaluation are some of the processes that make up the suggested framework. Before being sent to the chosen deep learning architectures, MRI images are first preprocessed and normalized. Each model predicts the relevant class labels after learning discriminative characteristics linked to Alzheimer's disease stages.

To enable thorough comparison analysis, both CNN-based and transformer-based models are included in the chosen architectures. The models that were assessed are:

- MobileNetV2
- EfficientNet-B0
- ResNet50
- DenseNet121
- Vision Transformer (ViT)

4.2 MobileNetV2

A lightweight convolutional neural network architecture called MobileNetV2 was created for effective computation and deployment in settings with limited resources. To lower computational complexity and parameter count, the model makes use of inverted residual blocks and depthwise separable convolutions.

The conventional convolution procedure is split into two parts by depthwise separable convolution:

- Depthwise convolution
- Pointwise convolution

This preserves competitive classification performance while drastically lowering the amount of calculations.

4.3 EfficientNet-B0

Compound scaling, which consistently scales network depth, breadth, and picture resolution, is the foundation of EfficientNet-B0. EfficientNet methodically balances these aspects to increase performance while lowering computational cost, in contrast to conventional scaling techniques.

The following is a representation of the compound scaling approach:

$$Depth = \alpha^{\phi} \tag{1}$$

$$Width = \beta^{\phi} \tag{2}$$

$$Resolution = \gamma^{\phi} \tag{3}$$

where α, β, and γ are scaling coefficients and ϕ controls the overall scaling factor.

4.4 ResNet50

A deep convolutional neural network called ResNet50 uses skip connections to implement residual learning. As network depth grows, deep neural networks frequently experience vanishing gradient issues. During training, residual connections facilitate effective gradient propagation.

The definition of the residual learning mechanism is:

$$y = F(x) + x \tag{4}$$

where $F(x)$ represents the residual mapping and x denotes the input feature map.

ResNet50 can learn intricate hierarchical feature representations from MRI scans thanks to its 50 layers.

4.5 DenseNet121

Each layer receives feature maps from every layer before it thanks to DenseNet121's introduction of dense connection between levels. The network's gradient flow and feature reuse are both improved by this architecture.

The expression for the dense connectivity mechanism is:

$$x_l = H_l([x_0, x_1, x_2, ..., x_{l-1}]) \tag{5}$$

where x_l represents the output of the current layer and $[x_0, x_1, ..., x_{l-1}]$ denotes concatenated outputs from previous layers.

By recycling learnt features, DenseNet121 lowers parameter redundancy and enhances classification performance.

4.6 Vision Transformer (ViT)

A transformer-based architecture called Vision Transformer (ViT) was created for image categorization applications. ViT separates an image into fixed-size patches and analyzes them as sequential tokens, in contrast to CNNs that rely on convolution operations.

To maintain spatial information, each picture patch is mixed with positional embeddings and linearly embedded. After that, the embedded patch sequence is run through many transformer encoder blocks.

4.7 Lightweight Vision Transformer Configuration

A lightweight Vision Transformer (ViT) design was used to increase computational performance and enable fair comparison with CNN-based systems. In comparison to conventional ViT-Base designs, the model uses fewer transformer encoder layers and smaller embedding dimensions. For the diagnosis of Alzheimer's disease using MRI scans, this method maintains high classification performance while drastically reducing the number of parameters.

4.8 Transformer Encoder Block

A Multi-Head Self-Attention (MHSA) layer and a position-wise feed-forward neural network make up the Transformer Encoder Block. By projecting each input token into query, key, and value representations, the MHSA technique allows the model to capture global contextual linkages and long-range dependencies.

To learn various feature representations from various subspaces, several attention heads work in simultaneously. While residual connections and layer normalization enhance training stability and gradient flow, positional embeddings maintain spatial information. Figure 3 illustrates the architecture of the Transformer Encoder Block.

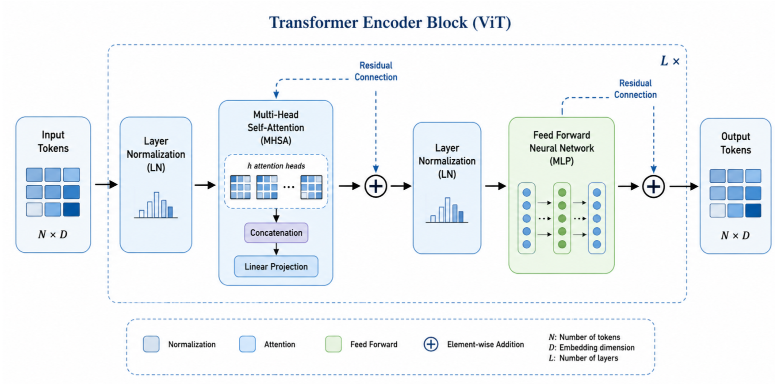

Fig. 3. Architecture of the Transformer Encoder Block used in Vision Transformer.

4.9 Training Strategy

To guarantee a fair comparison, all models were trained under the same experimental conditions.

The setup for training consists of:

- Optimizer: Adam

- Learning Rate: 1×10^{-4}
- Batch Size: 32
- Number of Epochs: 50
- Loss Function: Categorical Cross-Entropy

The Adam optimizer was chosen because of its effective convergence behavior and capacity for adaptive learning.

4.10 Loss Function

The loss function for multi-class classification problems was categorical cross-entropy. The definition of the loss function is:

$$L = -\sum_{i=1}^{C} y_i \log(\hat{y}_i) \tag{6}$$

where C represents the number of classes, y_i denotes the true label, and $\hat{y}_i$ represents the predicted probability.

4.11 Regularization Techniques

Several regularization strategies were used to reduce overfitting and enhance model generalization:

- Data augmentation
- Dropout layers
- Early stopping

Early stopping stops training when performance stops improving and keeps track of validity loss during training.

4.12 Performance Assessment Criteria

Several statistical metrics were taken into consideration in order to assess the efficacy of the suggested models. These measurements offer a thorough evaluation of categorization skill throughout various stages of Alzheimer's disease.

The overall percentage of properly identified samples is measured by accuracy:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \tag{7}$$

Precision evaluates the reliability of positive predictions:

$$\text{Precision} = \frac{TP}{TP + FP} \tag{8}$$

Recall measures the ability of the model to correctly identify positive samples:

$$\text{Recall} = \frac{TP}{TP + FN} \tag{9}$$

The F1-score combines precision and recall into a single performance metric:

$$F1 = \frac{2 \times Precision \times Recall}{Precision + Recall} \tag{10}$$

4.13 Implementation Environment

TensorFlow and Keras libraries were used in Python to create the suggested framework. To speed up calculation and shorten training time, training and assessment were carried out in a GPU-enabled environment. A GPU-enabled workstation with an NVIDIA GPU, an Intel Core i7 CPU, and 16 GB of RAM was used for the research. To guarantee a fair comparison of performance and computational efficiency, all models were trained and assessed using identical hardware and software.

5 Experimental Results and Discussion

5.1 Experimental Setup

TensorFlow and Keras frameworks were used in a GPU-enabled environment for the research. To establish a fair comparison, all of the chosen models were trained under the same circumstances.

The following is a summary of the training configuration:

- Optimizer: Adam
- Learning Rate: 1×10^{-4}
- Batch Size: 32
- Epochs: 50
- Image Size: 224×224

An 80:10:10 ratio was used to split the dataset into subgroups for testing, validation, and training.

5.2 Performance Comparison

Out of all the designs that were assessed, the Vision Transformer had the best categorization accuracy. Because of their deep feature extraction capabilities, DenseNet121 and ResNet50 also showed excellent classification performance.

The categorization accuracy attained by the assessed designs is shown in Figure 4. With an accuracy of 97.1%, the Vision Transformer outperformed DenseNet121 with 96.0%. The trade-off between predictive performance and resource needs was highlighted by MobileNetV2, which demonstrated the lowest accuracy while maintaining higher computing efficiency.

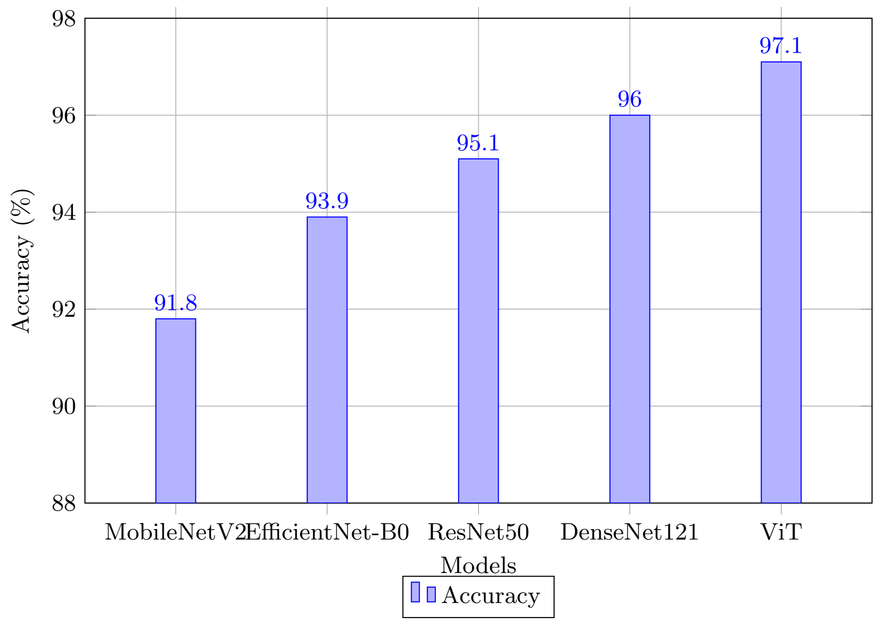

Fig. 4. Accuracy comparison of CNN and Vision Transformer architectures.

5.3 Efficiency Comparison

Computational efficiency was assessed utilizing parameter count, FLOPs, inference time, and model size in addition to classification results.

Table 1. Efficiency comparison of CNN and Vision Transformer architectures

Model	Parameters (M)	FLOPs (G)	Inference Time (ms)	Model Size (MB)
MobileNetV2	3.5	0.30	5	14
EfficientNet-B0	5.3	0.39	8	21
ResNet50	23.6	4.10	15	94
DenseNet121	8.0	2.90	12	32
Vision Transformer	8.7	5.20	22	35

Because MobileNetV2 had the shortest inference time and the lowest computing cost, it may be used in healthcare settings with limited resources. Vision Transformer, on the other hand, needed more processing power but produced the best categorization accuracy.

5.4 Confusion Matrix Analysis

Comprehensive information on class-wise classification performance is provided by the confusion matrix.

Table 2. Confusion Matrix Analysis of CNN and Vision Transformer architectures

Model	Precision	Recall	F1-Score
MobileNetV2	91.0	90.7	90.8
EfficientNet-B0	93.2	93.1	93.1
ResNet50	94.5	94.4	94.4
DenseNet121	95.4	95.3	95.3
Vision Transformer	96.5	96.4	96.4

According to the confusion matrix, the majority of MRI pictures were accurately categorized into the appropriate phases of Alzheimer's disease. Subtle structural abnormalities in the early stages of the disease led to minor misclassifications between Very Mild Dementia and Mild Dementia.

5.5 Attention Map Visualization

The Vision Transformer paradigm was used for attention-based visualization in order to enhance interpretability.

Figure 5 presents the attention map that highlights important brain regions associated with the classification of Alzheimer's disease.

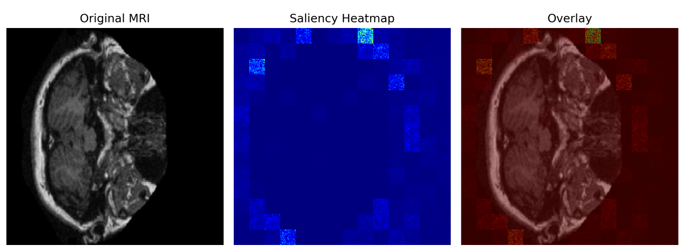

Fig. 5. Attention map visualization highlighting important brain regions for Alzheimer's disease classification.

The attention map shows that the model concentrates on brain regions like the hippocampus, ventricles, and cortical areas that are clinically significant. This bolsters the model's potential application in clinical decision support systems and increases trust in its predictions.

5.6 Comparative Analysis

Each architecture has unique advantages and disadvantages, as the comparative analysis shows. Because MobileNetV2 obtained the quickest inference speed and the lowest computing cost, it may be used in contexts with limited resources.

EfficientNet-B0 offered a good trade-off between efficiency and classification performance.

Because of their enhanced gradient propagation mechanisms and deeper feature extraction capabilities, ResNet50 and DenseNet121 were able to attain greater classification accuracy. Dense connection was especially helpful for DenseNet121, which allowed for effective feature reuse and improved learning of MRI-based illness patterns.

The Vision Transformer outperformed all other studied topologies in terms of categorization. Compared to traditional CNN-based models, the self-attention mechanism allowed the model to better capture global contextual information and long-range spatial linkages. These results imply that transformer-based designs have great promise for clinical decision support applications and automated Alzheimer's disease detection.

5.7 Ablation Study

The effect of augmentation and regularization methods on model performance was assessed by an ablation research.

Table 3. Ablation study results

Experiment	Accuracy (%)
Without Augmentation	93.2
With Augmentation	97.0
Without Dropout	94.1
With Dropout	97.0

The results indicate that data augmentation and dropout regularization significantly improve model generalization and classification performance.

5.8 Discussion

The study's conclusions emphasize the trade-off between computing efficiency and classification accuracy. Lightweight CNN designs provide quicker inference and lower computing needs, even if Vision Transformer performed better in classification.

Due to minute structural variations across closely related classes, early-stage Alzheimer's disease categorization is still difficult. However, the suggested framework showed a great capacity to detect the course of the disease from MRI pictures.

Overall, the findings show that transformer-based designs offer great potential for clinical decision support and automated medical picture processing.

5.9 Limitations

Even with the positive outcomes, there are still a few restrictions. The study used 2D MRI slices, which might not have captured all of the volumetric data included in 3D brain scans. Additionally, model generalization may be impacted by dataset-specific features and class imbalance. To increase robustness and clinical usefulness, future research should assess the suggested framework using bigger multi-center datasets and integrate multimodal imaging modalities.

6 Conclusion

In this work, a detailed comparative study was given to assess the performance of CNN and Vision Transformer architectures for multi-class Alzheimer's disease classification using MRI images. To ensure a fair comparison, the same experimental conditions were used to evaluate the models, which were selected as MobileNetV2, EfficientNet-B0, ResNet50, DenseNet121, and Vision Transformer.

The experimental results indicated that the Vision Transformer performed the best in terms of classification performance among all the architectures that were evaluated. The self-attention mechanism allowed the model to learn global contextual relationships and long-range dependencies more efficiently than traditional CNN-based architectures. DenseNet121 and ResNet50 also showed good classification ability due to their deep hierarchical feature extraction mechanisms.

Lightweight architectures like MobileNetV2 resulted in reduced computational complexity, inference time and model size making them suitable for deployment in resource-constrained healthcare settings. The comparative analysis demonstrated a trade-off between classification accuracy and computational efficiency across architectures.

The attention-based visualization also revealed that the Vision Transformer attended to the brain regions clinically relevant to Alzheimer's disease progression, such as hippocampus, ventricles, and cortical areas. This increase in interpretability increases confidence in the model predictions and allows it to be used in clinical decision support systems.

The significance of augmentation and regularization strategies in enhancing model generalization and classification performance was validated by the ablation research. Even with the encouraging outcomes, there are still several drawbacks, such as the restricted diversity of datasets and the use of 2D MRI slices.

In order to obtain richer volumetric data, future research will concentrate on expanding the suggested framework to 3D MRI analysis. Multimodal learning combining MRI, PET scans, and clinical data might be further enhancements. To increase clinical applicability and deployment efficiency, more study can investigate explainable AI methodologies, lightweight optimization strategies, and hybrid CNN-transformer architectures.

Overall, the results of this work show that Vision Transformer-based designs offer great promise for assisting automated diagnosis and intelligent healthcare systems and are very successful for multi-class Alzheimer's disease categorization.

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